

could also be used to monitor environmental factors such as earthquake aftershocks and tsunami second waves.

The new portable system could help save lives by allowing basic systems to be put in place to plug the gap during the first 24-48 hours following a disaster and allow communication with emergency services and survivors until full-scale systems can be restored.

First responders would be able to transport the system in a single box or briefcase to the site and instantly set up communication systems that have been damaged or eliminated during the disaster. Inbuilt battery power supplies fitted to briefcases or transportation boxes would provide power for up to two days, while solar panels could also be used in areas of good lighting.

Fastest phased-array 5G communication link

Keysight Technologies and the University of California (UC) San Diego have demonstrated the world's fastest bidirectional phased-array link in the 28GHz 5G band. The achievement is an important milestone for delivering future applications in 5G, aerospace and defence.



University of California San Diego's demonstration of wideband, 28GHz, pre-5G phased-array signal transmission, with the help of a Keysight 5G millimetre-wave testbed (Image courtesy: Keysight Technologies, Inc.)

The demonstration included a 64-element array that achieved a data rate of 12Gbps at 0 degree and more than 8 Gbps over all scan angles up to ± 50 degrees in azimuth and ± 25 degrees in elevation at a link distance of 300 metres. The array produced data rates of up to 18Gbps at shorter distances. The results did not rely on any calibration on the 64-element phased-array, thereby greatly reducing implementation costs.

The 64-element phased array, built on a low-cost printed circuit board, consumed only 7-11W of DC

power in either its transmit or receive mode, thanks to the UC San Diego high-performance system-on-a-chip (SoC) design that uses a third-generation silicon germanium process.

UC San Diego used Keysight's 5G millimetre-wave testbed for the communication link.

Combination of features produces new Android vulnerability

A new vulnerability affecting Android mobile devices results not from a traditional bug but from the malicious combination of two legitimate permissions that power desirable and commonly-used features in popular apps. The combination could result in a new class of attacks, which has been dubbed 'Cloak and Dagger.'

The vulnerability, which was identified and tested in closed environments by computer scientists at the Georgia Institute of Technology, would allow attackers to silently take control of a mobile device, overlaying its graphical interface with false information to hide malicious activities being performed underneath—such as capturing passwords or extracting the user's contacts. A successful attack would require the user to first install a type of malware that could be hidden in a pirated game or other app.

The issue could be more difficult to resolve than ordinary operating system bugs. Said Wenke Lee, a professor in Georgia Tech's School of Computer Science and co-director of the Institute for Information Security & Privacy, "The two features involved are very useful in mapping, chat or password manager apps, so preventing their misuse will require users to trade convenience for security. This is as dangerous an attack as we could possibly describe."

The first permission feature involved in the attack supports the use of devices by disabled persons, allowing inputs such as user name and password to be made by voice command, and allowing outputs such as a screen reader to help the disabled view content. The second permission is an overlay or 'draw on top' feature that produces a window on top of the device's usual screen to display bubbles for a chat program or maps for a ride-sharing app.

When combined in a malicious way, the accessibility permission acts as a cloak, while overlay serves as the dagger. The two could allow attackers to draw a window that fools users into believing they are interacting with legitimate features of the app. The malicious program, operating as the overlay, would then capture the user's credentials for the malware author, while the accessibility permission would enter the credentials into the real app hidden beneath, allowing it to operate as expected and leaving the user with no clue that anything is awry.